

Climate, Environment, and Equity Impacts of CEMEX Air Emissions

Greenhouse Gas Emissions from the CEMEX Facility

The climate crisis is affecting all of Boulder County through heat, drought, unreliable and extreme precipitation, wildland and urban firestorms, economic and agricultural insecurity. These issues have disproportionate impacts on income-constrained, housing-and energy-cost-burdened residents, youth and senior residents, and residents of color.

Lyons itself is located in a semiarid ecosystem of historically fire-mediated grassland at the confluence of major rivers, and so it is at particular risk from drought, fire, extreme precipitation and flooding. The likelihood of flooding, and consequent impacts to water quality downstream, is exacerbated by fires in the mountains upstream. Lyons was devastated by the 2013 flood, which [climate experts have found](#) was made much more likely by climate change. The Lyons flood disproportionately destroyed affordable housing, including nearly 50 mobile homes. Many [low-income residents](#) who were displaced by that climate-related disaster have not been able to return to their community. This displacement, in turn, puts additional pressure on affordable housing availability throughout the county and increases the risk of financial insecurity and homelessness.

CEMEX is the only major industrial source of greenhouse gas emissions (GHG) in Boulder County. As reflected in [Boulder County's GHG inventory](#), these emissions constitute the single largest point source of GHG emissions in the county. The GHG emissions from this plant are largely carbon dioxide from the combustion of natural gas and coal in a kiln that produces clinker, the basic component of cement used in construction materials including concrete.

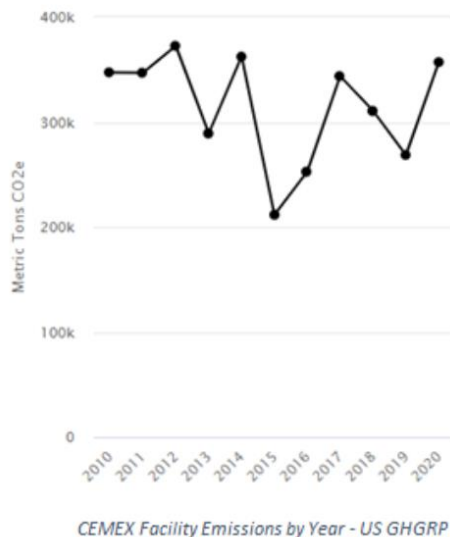
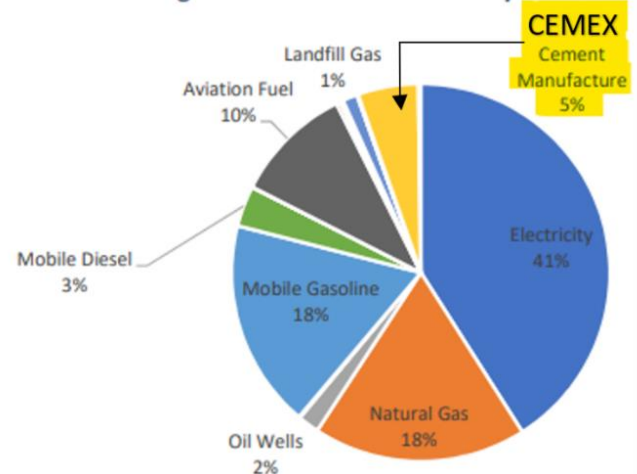


Figure ES- 2. 2016 Emissions by Source



Boulder County Greenhouse Gas Inventory 2016

CEMEX' greenhouse gas emissions are accounted for in the [EPA's Greenhouse Gas Reporting Program \(GHGRP\)](#), to which the company is required to report annual emissions determined by the difference between feedstocks entering the process and carbon exiting the process. Since the outset of this data reporting in 2010, emissions of methane and nitrous oxide have decreased as some controls have been put into place. However, recent total GHG emissions have returned to well-above the average of all reported years since 2010.

The impact of greenhouse gases, particularly of CEMEX' major GHG carbon dioxide, is felt cumulatively over time as their concentration builds up in the atmosphere. Thus, it's important to recognize that while we only have GHG data going back to 2010, the plant's main kiln has been in operation since 1977 with fewer emissions controls in early years.

	Carbon Dioxide (metric tons CO₂e)	Methane (CH₄) (metric tons CO₂e)	Nitrous Oxide (N₂O) (metric tons CO₂e)	Total Emissions (metric tons CO₂e)
2010	346,178	433	749	347,360
2011	345,675	473	817	346,965
2012	371,723	353	596	372,672
2013	288,093	405	698	289,196
2014	360,911	516	890	362,317
2015	211,024	243	415	211,682
2016	252,139	212	354	252,705
2017	343,201	220	362	343,783
2018	310,575	4	4	310,583
2019	268,639	2	2	268,643
2020	357,095	3	3	357,101
<i>(cited in CEMEX August 2022 GEMM phase 1 audit report; not yet in EPA FLIGHT):</i>				
2021	292,006	325	545	292,876

CEMEX Facility Emissions by Year, Type - US GHGRP

The above emissions do not include Scope 3 emissions that are indirect emissions from the extraction and production of the purchased input materials to the cement process, fuels, transport-related activities in vehicles not operated by this location and other outsourced activities such as waste disposal. In terms of incorporating all the emissions relative to cement including scope 3, this [analysis](#) discusses *all* emissions related to the cement manufacturing process.

CEMEX as an international company has made inroads (with federal assistance) into [process efficiencies as well as lower-carbon cement](#). However these improvements have not materialized at the Lyons plant to date.

The primary policy currently acting to specifically reduce GHG emissions from CEMEX is the new [Greenhouse Gas Emissions and Energy Management for Manufacturing in Colorado \(GEMM\) Phase 1](#) rule governing “energy-intensive, trade-exposed” (EITE) emissions sources including cement and steel plants. Only four facilities in Colorado fall into this category – three in Pueblo and one in Lyons. Three are cement manufacturers. Finalized in 2021, the GEMM phase 1 rule requires CEMEX to complete an audit of most of its emitting equipment that will make recommendations for upgrading the facility for GHG emissions and recommend which upgrades of comparable cost-effective offer greater “co-benefits,” where “co-benefit” refers to the reduction of other air pollutants, such as air toxics, that would benefit nearby communities. Starting in 2025, the facility must demonstrate GHG emissions control and energy efficiency best practices plus 5% less GHG emissions by mass than a calculated limit that allows for increase in production activity. CEMEX status as EITE is the reason for the allowed

production increase and the much lower reduction requirement (the entire manufacturing sector is required by law to reduce GHG emissions 20% by 2030). While the rule requires a facility to prioritize “GHG reduction initiatives that would produce co-benefits to the neighboring communities surrounding the EITE stationary source,” that is only when comparing “compliance options with similar or the same cost-effectiveness.” There is additional significant leniency as to which of the audit-analyzed upgrades a facility must complete based on cost, feasibility, and competitiveness for the facility. The audit plan was posted [here](#) in August and the audit results and recommendations are due December 31, 2022. The audit plan is specific that only the cement kiln will be included in the audit scope; it is not specific as to what type and depth of retrofit is likely to be recommended, though it cites the GEMM phase 1 rule requirement to evaluate carbon capture and utilization or storage.¹

With as little as 5% GHG emissions reductions and the flexibility afforded by CEMEX’s EITE status, there is no guarantee the GEMM phase 1 will result in near-term GHG emissions reductions from the Lyons facility on the scale implied by Boulder County’s greenhouse gas reduction goals. [Boulder County’s greenhouse gas reduction goals](#) are tuned to the reductions deemed necessary by the Intergovernmental Panel on Climate Change (IPCC) to limit global warming to 1.5 °C above pre-industrial levels. Accordingly, Boulder County as a whole must reduce emissions at least 45% by 2030, and therefore, the Lyons plant will need to aggressively reduce their impact to accomplish their share of this transition.²

Conclusion

This is a crucial moment. The latest IPCC report warns that the window of opportunity to avert climate catastrophe is rapidly shrinking and limiting average warming to 1.5°C over the next 10 years is critical to prevent the worst effects of climate change. Globally we have already reached 1.1°C of temperature increases, and without immediate reductions in climate pollution it is likely Earth will exceed that degree of warming at least once this decade.

It is a challenge to determine the long-term environmental, climate, and human health impacts caused by the plant if it were to run through this crucial decade or indefinitely. As mentioned above, it’s noteworthy that CEMEX (as a parent company) has aggressive climate goals. CEMEX USA President Jaime Muguero shares “CEMEX is committed to being part of the solution to reduce carbon emissions globally and to deliver net-zero CO₂ concrete to all of our customers by 2050” and they are looking for public-private partnerships to deploy technology at their locations to ensure they operate in a way that can achieve these goals. [Here](#) is what they are currently doing with funding from the Department of

¹ CCR 1001-26 Air Quality Control Commission:

“II.C.3.a.(i)(A)(1) Control technologies and strategies [for the GHG BAECT analysis] must include, but are not limited to, fuel use, raw material use, energy efficiency improvements, preheating/heat reuse and strategic energy management options), and carbon capture and underground storage or utilization

II.C.3.a.(i)(B) Eliminate technically infeasible control technologies and strategies

II.C.3.a.(i)(B)(1) Notwithstanding Section II.C.3.a.(i)(B), the audit team must perform a feasibility assessment of carbon capture and underground storage or utilization technology for any single emissions unit evaluated with direct emissions of 100,000 tons per year or greater CO₂e in any of the previous five years as reported under Regulation number 22, Part A and/or 40 CFR Part 98. The audit team shall include this analysis in the audit report.”

² Boulder County is currently updating its climate goals to reflect the science-based targets needed to achieve a livable planet; thus, we will need to get even more emissions reductions in order to achieve these new climate goals (to be adopted in December 2022 pending approval of Board of County Commissioners).

Energy. We see it as achievable and essential that the Lyons CEMEX plant follow suit and commit to decarbonizing its operations as rapidly as possible.

As discussed above, CEMEX will be required by state law to make adjustments to its future operations that will reduce the plant's impact. However, so long as plant operations comply with state law, operations will be allowed to continue indefinitely. Likewise, CEMEX has asserted that it will continue to operate the plant even if the County denies the special use permit at issue. From an environmental health and climate perspective, securing a definitive future closure date is better than allowing the operations to continue indefinitely. Even with a definitive future closing date, Boulder County staff should work toward securing considerable reductions in its impact during that time period, including working on state legislative and regulatory action to ensure the plant significantly reduces their greenhouse gas impacts.

Allowing the operations of the plant to continue for any timeframe is a concern locally. If the plant ceases operations, it is possible this action would result in a transfer of the burden to another community where new cement operations would start or increase, perhaps under less stringent GHG or air pollution regulations, to the detriment of local environmental justice or net CO₂ emissions, so it is critical that CEMEX maintain its commitment to next-generation net-zero CO₂ concrete. Given this is the largest stationary source of greenhouse gas emissions by far in Boulder County, and that Boulder County has science-aligned climate goals, it's necessary to these goals that CEMEX in Lyons either become part of the solution to reduce its current impact as much as possible, and eventually transition to deliver net-zero CO₂ concrete, or to cease operations altogether in the near future.

Other Air Pollutant Emissions from the CEMEX Facility

The CEMEX plant also operates under an Operating Permit, Permit 95OPBO082, issued by the Colorado Department of Public Health and Environment's Air Pollution Control Division. This permit addresses emissions from the plant of "criteria air pollutants" identified as such under the Clean Air Act -- nitrogen oxides (NO_x), sulfur oxides (SO_x), carbon monoxide (CO), volatile organic compounds (VOCs), and particulate matter (PM). NO_x and VOCs combine with sunlight to produce ozone, another criteria pollutant. Emissions of these pollutants are limited by the U.S. Environmental Protection Agency (EPA) based on their health impacts. Boulder County is part of the Denver Metro/North Front Range nonattainment area for ozone. This means that this area does not meet -- is out of attainment with -- EPA limits for ozone exposure. The area has been out of attainment for many years, and EPA is in the process of further downgrading the attainment status to "severe" nonattainment with the 2008 federal ozone standard.

Although the Colorado Air Pollution Control Division (APCD) is responsible for issuing, revising and renewing CEMEX's air permit, EPA also reviews the permit and has enforcement authority. The air permit limits pollution from all of the air emissions sources in the facility. The facility overall has 22 different regulated emission sources, including emissions from the Dowe Flats Quarry. CEMEX reports its annual emissions to APCD and those are included in annual inspection reports.

At the Dowe Flats Quarry, the air permit regulates the amount of total material handled and processed, and air emissions from blasting, other quarry activities, the primary crusher at the quarry, and the belt conveyor. Air emissions from blasting and fugitive emissions from other quarry activities are not controlled by air pollution control equipment, but rather limited by the amount of material handled. Fugitive particulate, a common subject of local concern at the quarry, is particulate matter that has not

passed through a stack (such as a chimneys, pipe, vent, or duct) before being released to the air. "Fugitive dust" is a type of fugitive particulate matter released into the air by wind or other similar forces.

Emissions from the primary crusher at the quarry and the belt conveyor are controlled by baghouses, which capture and reduce particulate matter (PM) emissions from those sources.

The primary pollutant coming from the Dowe Flats Quarry activity is PM and particulate matter with particles less than 10 micrograms in size (PM₁₀). [According to the EPA](#), PM₁₀ is the greater health concern, because these particles can get deeper into a person's respiratory and circulatory system. EPA states that exposure to particulates can affect people's lungs and heart. Studies have linked particulate pollution to:

- Premature death in people with heart or lung disease
- Nonfatal heart attacks
- Irregular heartbeat
- Aggravated asthma
- Decreased lung function
- Increased respiratory symptoms, such as irritation of the airways, coughing or difficulty breathing.

From an environmental perspective, PM pollution can reduce visibility, settle on water or land and make lakes and streams more acidic, deplete nutrients in soil, damage sensitive forests and farm crops, and affect the diversity of ecosystems.

Reviewing annual inspection report data from CEMEX for 2015 – 2020 (excluding 2019, when an annual inspection was not performed), the average emissions of (PM and the smaller PM₁₀ from the Dowe Flats Quarry were as follows:

Blasting and fugitive PM emissions (in tons per year (TPY)) from quarry activities (AIRS ID 017 and 025)

Pollutant	Permit Limit (TPY)	Average reported emissions 2015-2020 (TPY)
PM	134.2	96.04
PM ₁₀	58.4	29.98

Note that Cemex began tracking NO_x and CO emissions from blasting activities in 2018. The permit limits NO_x from blasting to 10 tons per year (tpy), and CO to 39.6 tpy. Reported emissions were less than a ton for each pollutant for 2018 and 2020.

Primary Crusher, Belt Conveyor, Radial Stacker to Stockpiles (AIRS ID 027 and 026)

Pollutant	Permit Limit (TPY)	Average reported emissions 2015-2020 (TPY)
PM	0.16	0.0528
PM ₁₀	0.07	0.02284

With respect to emissions at the rest of the CEMEX facility, the kiln is by far the largest emission source of other criteria pollutants. Reviewing the reported emissions data included in the 2015-2020 inspection reports, the average emissions from the kiln were as follows:

Criteria Pollutant Emissions from Kiln Burning (AIRS ID 007))

Pollutant	Permit Limit (TPY)	Average reported emissions 2015-2020 (TPY)
PM	133	10.962
PM ₁₀	133	10.962
NO _x	901	287.41
SO _x	95	9.94
CO	396	166.73
VOC	138	3.53

CEMEX Compliance History

The Colorado Air Pollution Control Division provides regular permit compliance oversight of the CEMEX Lyons plant, including annual inspections, oversight of any required stack testing from emission points, and other air quality compliance related issues. Delegated representatives from Boulder County Public Health (BCPH) typically take the lead responding to complaints regarding the facility. According to APCD, CEMEX has been the subject of 12 formal enforcement actions by APCD since 2000, five of these in the last five years. In addition, there is a formal enforcement action by APCD still pending.

The majority of these APCD enforcement actions have related to violations of permit requirements related to air quality monitoring and testing at the kiln and on the Lyons site. Other bases for formal enforcement have been recordkeeping and reporting violations. There have also been several dust and opacity events from the kiln and the clinker cooler. The Dowe Flatts Quarry operations have not been sources of formal compliance enforcement.

The majority of complaints made to Boulder County Public Health have been related to fugitive dust issues from the plant side of operations, although there have been complaints about dust and seismic activity related to the Dowe Flatts Quarry over the years. The number of complaints varies from year to

year. For example, so far in 2022 there have been 4 complaints; in 2021 there were 15 complaints; in 2020 there were 4 complaints; and in 2019 there were 16 complaints related to the plant side of the operation. BCPH response involves investigating each complaint by contacting the company for an explanation of the reason for the event, requesting a plan to rectify the issue, site visit depending on circumstances, forwarding event details to APCD when necessary, and logging details of the event and response in the APCD database. In numerous meetings with CEMEX management and staff, BCPH has requested several upgrades and preventative measures be addressed at the Plant in response to these complaints.

APPENDIX: 2020 GHG SUMMARY REPORT FOR CEMEX LYONS

Available at <https://ghgdata.epa.gov/ghgp/main.do>

Facility Name:CEMEX CONSTRUCTION MATERIALS SOUTH LLC

Facility Identifier:

Facility Reporting Year:2020

Facility Location:

Address: 5134 UTE HIGHWAY

City: LYONS

State: CO

Postal Code: 80540

Facility Site Details:

CO2 equivalent emissions from facility subparts C-II, SS, and TT (metric tons):357,100.7

CO2 equivalent emissions from supplier subparts LL-QQ (metric tons):

Biogenic CO2 emissions from facility subparts C-II, SS, and TT (metric tons):0

Cogeneration Unit Emissions Indicator:N

GHG Report Start Date:2020-01-01

GHG Report End Date:2020-12-31

Description of Changes to Calculation Methodology:

Plant Code Indicator:N

Primary NAICS Code:327310

Second Primary NAICS Code:

Parent Company Details:

Parent Company Name:CEMEX INC

Address:10100 Katy Freeway, Suite 300, Houston, TX 77043

Percent Ownership Interest:100

Subpart C: General Stationary Fuel Combustion

Gas Information Details

Gas Name	Carbon Dioxide
Gas Quantity	5,220.5 (Metric Tons)
Own Result?	

Gas Name	Biogenic Carbon dioxide
Gas Quantity	0 (Metric Tons)
Own Result?	

Gas Name	Methane
Gas Quantity	0.1 (Metric Tons)
Own Result?	

Gas Name	Nitrous Oxide
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Gas Quantity	0.01 (Metric Tons)
Own Result?	

Unit Details:

Unit Name : GP-01

Unit Type : OCS (Other combustion source)

Unit Description :

Other Unit Name :

Small Unit Aggregation Details:

Use Ivt Indicator: N

Highest Maximum Rated Heat Input Capacity: 30

Cumulative Maximum Rated Heat Input Capacity: 30

Emission Details:

Annual CO₂ mass emissions from sorbent: 0 (Metric Tons)

Annual Biogenic CO₂ Emissions: 0 (metric tons)

Annual Fossil fuel based CO₂ Emissions: 5220.5 (metric tons)

Tier Fuel Details:

Fuel : Natural Gas (Weighted U.S. Average)

Tier Name : Tier 2 (Equation C-2a)

Tier Methodology Start Date : 2020-01-01

Tier Methodology End Date : 2020-12-31

Frequency of HHV determinations : Monthly

Tier 2 Monthly HHV Details :

January	February	March	April	May	June	July	August	September	October	November	December
Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Fuel Emission Details :

Total CO ₂ emissions	Total CH ₄ emissions	Total N ₂ O emissions	Total CH ₄ emissions CO ₂ e	Total N ₂ O emissions CO ₂ e
5220.5 (Metric Tons)	0.10 (Metric Tons)	0.010 (Metric Tons)	2.5 (Metric Tons)	2.9 (Metric Tons)

Equation C2a/C9a Inputs :

Fuel Quantity : 88934190 (scf/year)

Use Default High Heat Value : true

High Heat Value : 0.001106312 (mmBtu/scf)

Subpart H: Cement Production

CEMS Monitoring Location (CML) Details

CML Name : P007 Main Kiln

Description :

Configuration Type : Single process/process unit exhausts to dedicated stack

Types of Fuel Combusted :

Tier 4 Methodology Information

Methodology Start Date : 2020-01-01

Methodology End Date : 2020-12-31

Quarterly CO2 Emissions

First Quarter : 95606.7 (metric tons)

Second Quarter : 70689.79 (metric tons)

Third Quarter : 98652.39 (metric tons)

Fourth Quarter : 86925.79 (metric tons)

Annual CO2 Emissions

Total annual CO2 mass emissions (biogenic and non-biogenic) measured by the CEMS: 351874.7 (metric tons)

Emissions reported for the CEMS include emissions calculated according to 98.33(a)(4)(viii) for a slipstream that bypassed the CEMS. :

Total annual biogenic CO2 : 0 (metric tons)

Total annual non-biogenic CO2 : 351874.7 (metric tons)

Equation C-10 Summary and Results

Total CH4 : (metric tons)

Total N2O : (metric tons)

Additional Emissions Information

Total source operating hours in the reporting year	7152 (hours)
Total Operating hours in which a substitute data value was used in the emissions calculations for CO2 concentration	0 (hours)
Total Operating hours in which a substitute data value was used in the emissions calculations for stack gas flow rate	63 (hours)
Total Operating hours in which a substitute data value was used in the emissions calculations for stack gas moisture content	0 (hours)

Process Units:

Unit Name : P007 Kiln

Gas Information Details

Gas Name	Carbon Dioxide
Gas Quantity	351,874.7 (Metric Tons)
Own Result?	

Gas Name	Biogenic Carbon dioxide
Gas Quantity	0 (Metric Tons)
Own Result?	

Gas Name	Methane
Gas Quantity	0 (Metric Tons)
Own Result?	

Gas Name	Nitrous Oxide
Gas Quantity	0 (Metric Tons)
Own Result?	

Summary Information for the facility:

Number Of Kilns	1
Number Of Operating Kilns	1
Annual cement production for the facility	()
Annual clinker production	()
Annual average clinker CO2 emissions factor across all kilns	()
Annual average CKD CO2 emissions factor across all kilns	()

No Cems Cement Kiln Details:

Cems Cement Kiln Details:

Kiln Name : P007 Kiln

Kiln Type : Cement Kiln

Kiln Description :

Monthly Inputs/Outputs:

Month Name		
January		
	Clinker Production	()
	Clinker Production Indicator	N
February		
	Clinker Production	()
	Clinker Production Indicator	N
March		
	Clinker Production	()
	Clinker Production Indicator	N
April		
	Clinker Production	()
	Clinker Production Indicator	N
May		
	Clinker Production	()
	Clinker Production Indicator	N
June		
	Clinker Production	()
	Clinker Production Indicator	N
July		
	Clinker Production	()
	Clinker Production Indicator	N
August		
	Clinker Production	()

	Clinker Production Indicator	N
September		
	Clinker Production	()
	Clinker Production Indicator	N
October		
	Clinker Production	()
	Clinker Production Indicator	N
November		
	Clinker Production	()
	Clinker Production Indicator	N
December		
	Clinker Production	()
	Clinker Production Indicator	N